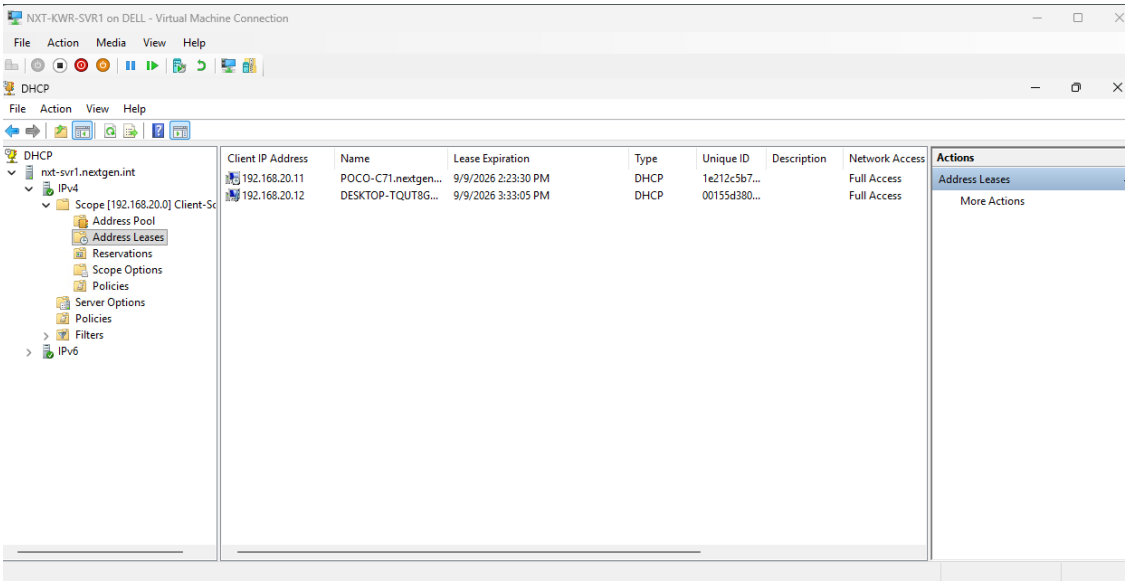
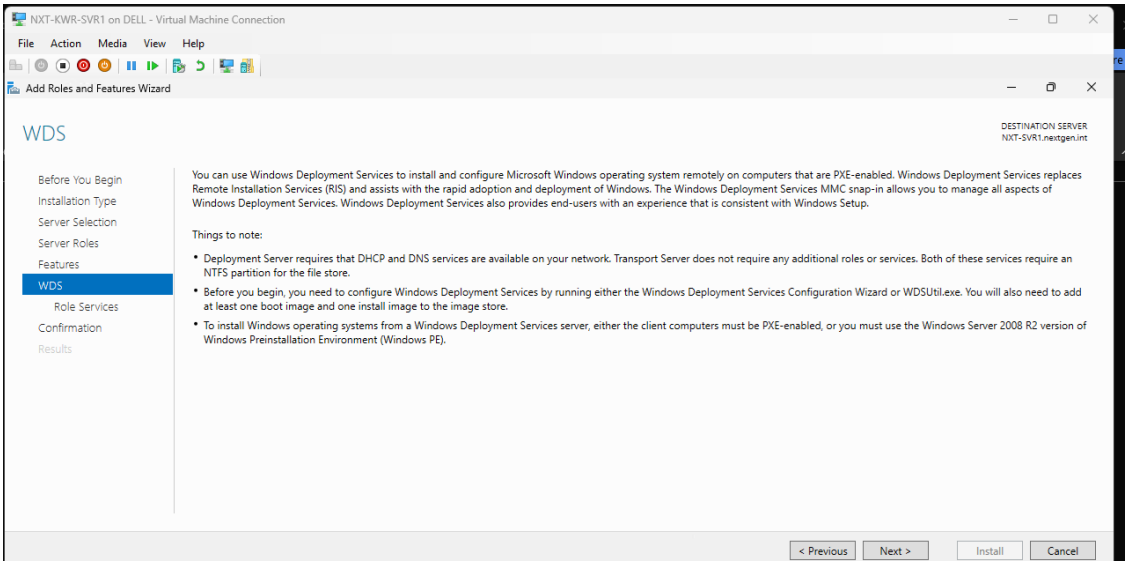


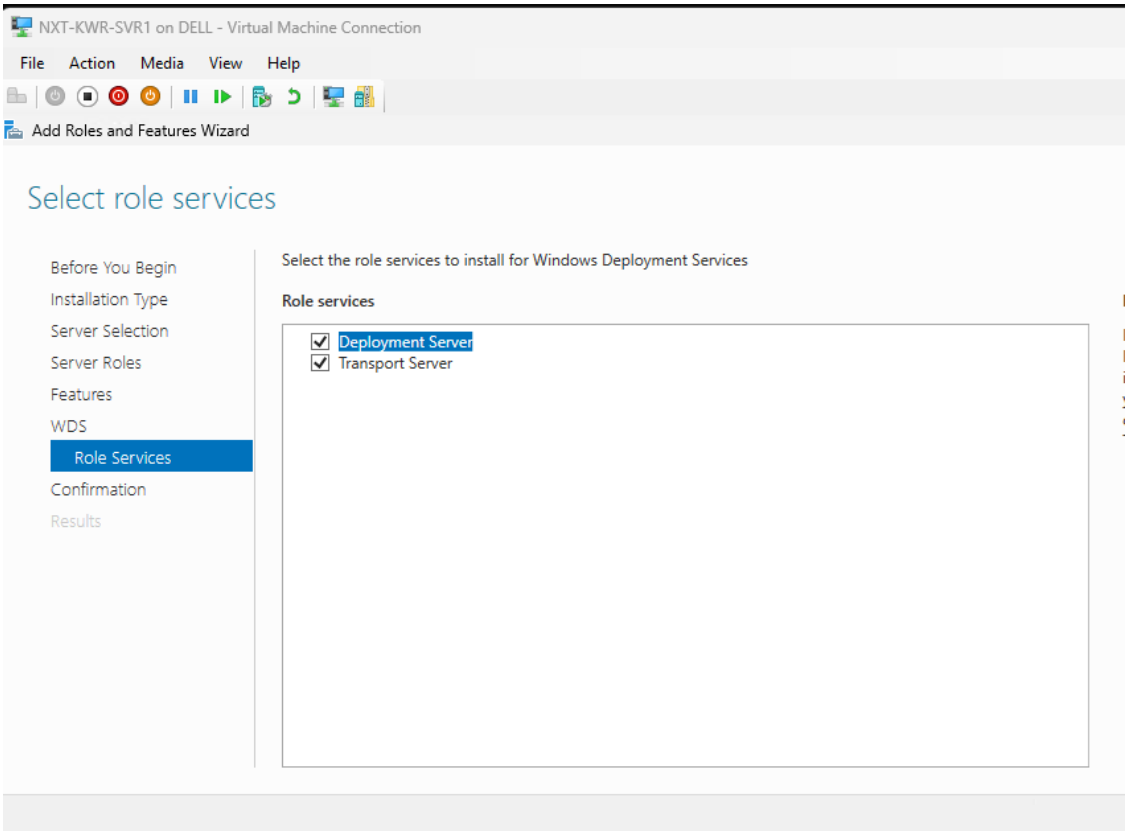
Part 2 — Windows Deployment Services (WDS)

With DHCP working, client computers can obtain an IP address on their own. The next step is deploying WDS, which is what actually delivers the Windows operating system over the network. In this lab, WDS is installed on a separate server, not on the Active Directory Domain Controller.



What WDS Requires

Windows Deployment Services allows administrators to deploy Windows operating systems over the network to PXE-enabled computers. Before configuring WDS, both DHCP and DNS must already be available on the network, since WDS depends on them for addressing and name resolution. WDS also requires an NTFS-formatted partition to store its image files. Once the role is installed, the administrator still needs to configure WDS and add at least one Boot Image and one Install Image before any client can be deployed to.

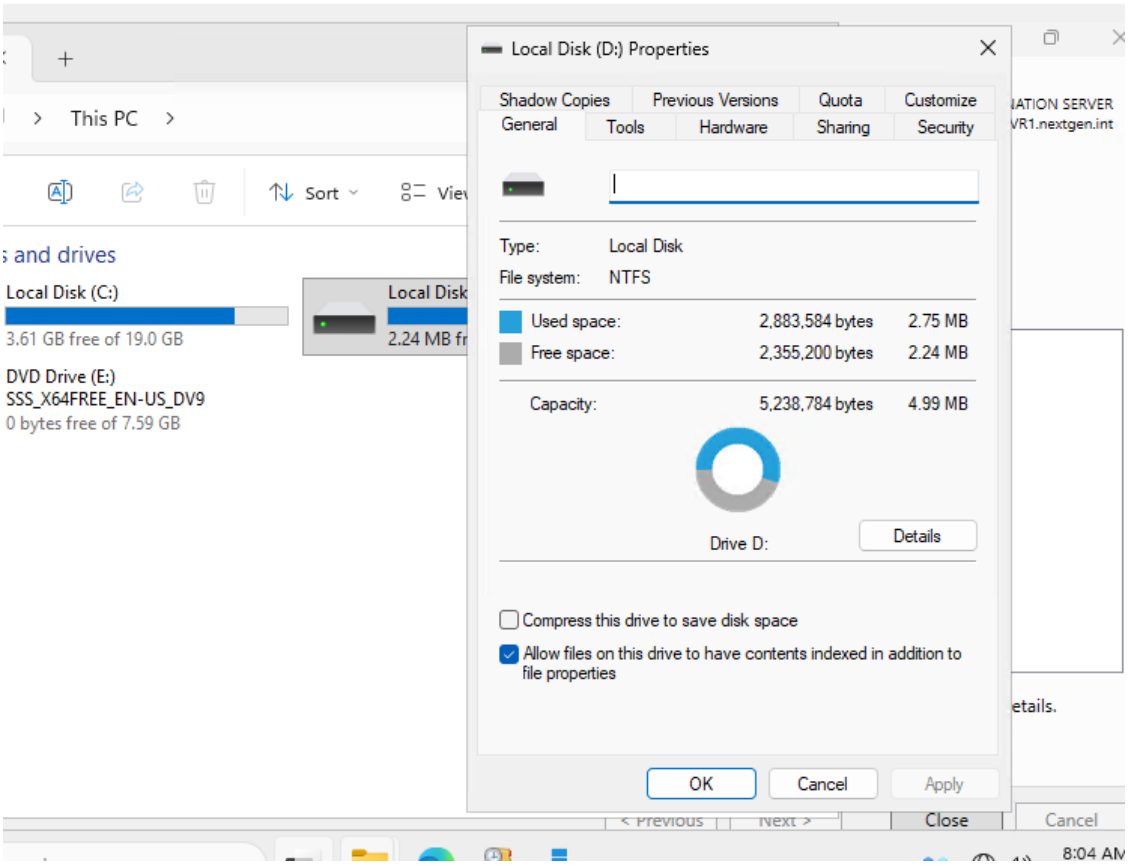


Deployment Server vs Transport Server

When installing the WDS role, you are offered two role services. Both are ticked in this lab, since we want full deployment functionality.

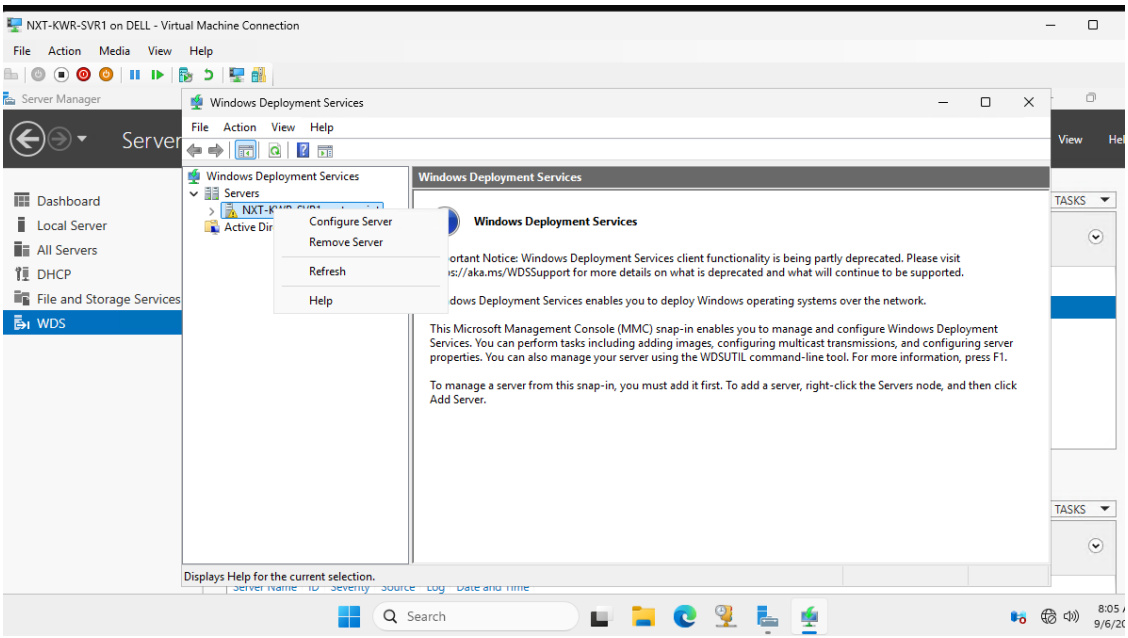
Role Service	What it does
Deployment Server	Provides the full WDS feature set: deploys Windows operating systems, manages boot and install images, and handles PXE-based installation over the network.
Transport Server	A stripped-down role that only transfers image data across the network. It does not support Windows deployment or PXE boot on its own.

As mentioned, WDS needs an NTFS partition available for storing its images — make sure one exists before continuing.

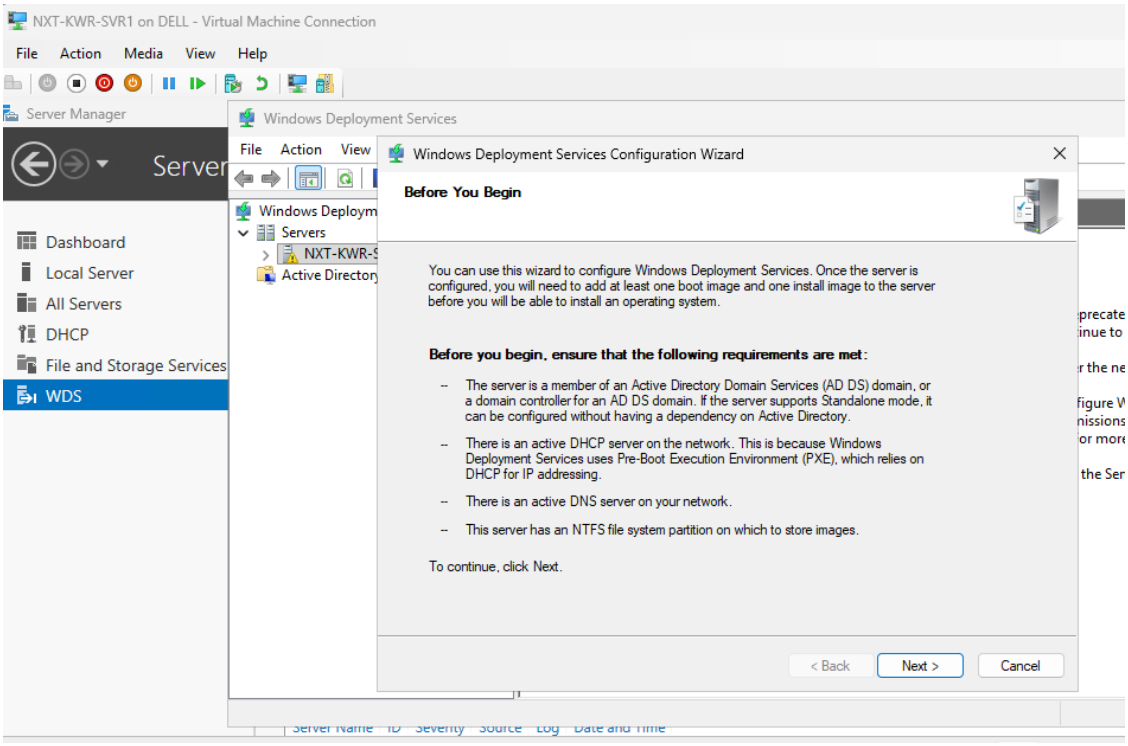


Opening and Configuring the WDS Server

Once installed, WDS can be opened as a shortcut from Windows Tools.

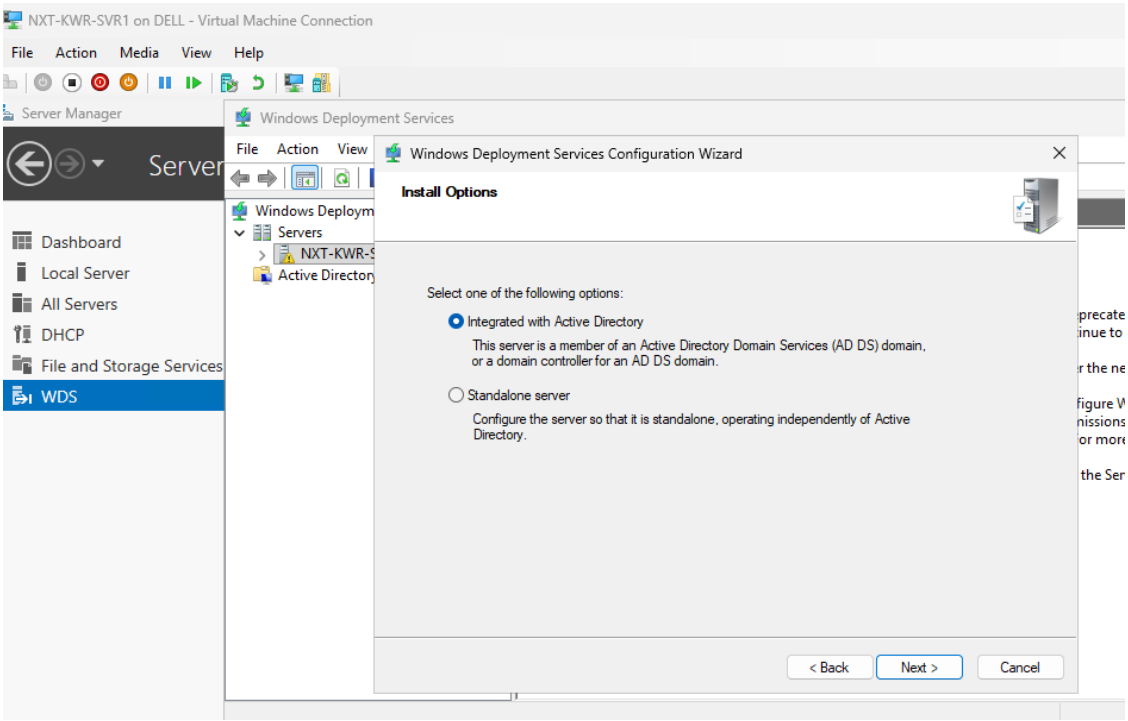


From there, right-click the server and choose Configure Server to launch the setup wizard.

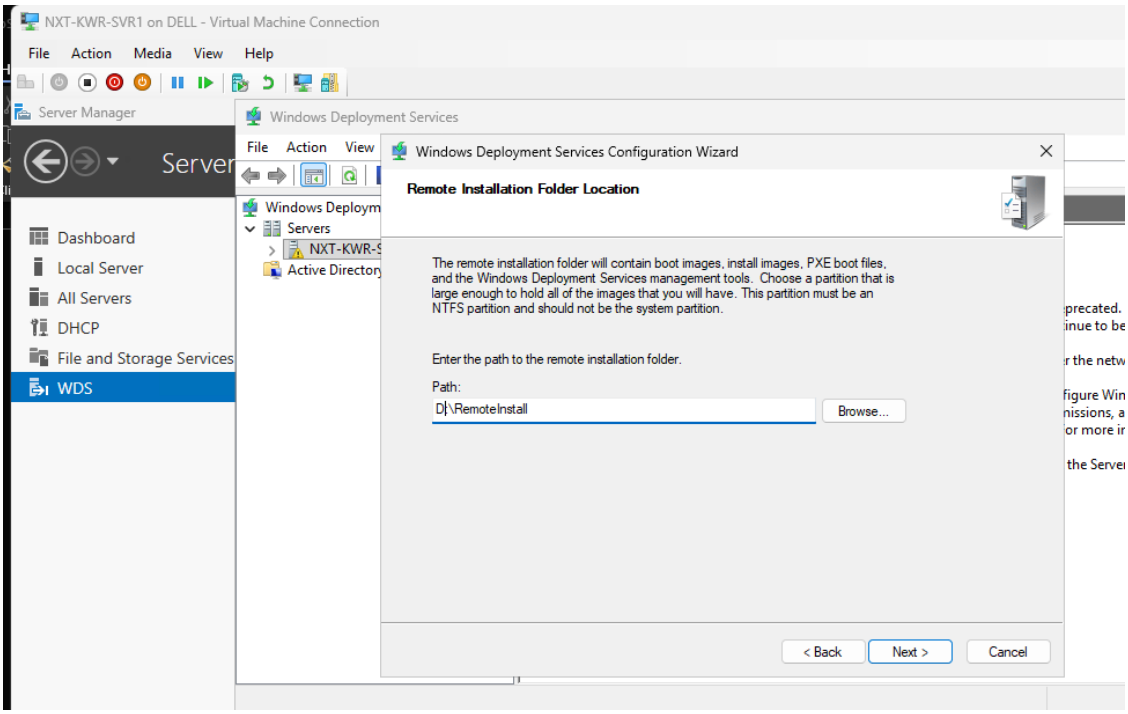


Prerequisites Check

Before configuring WDS, the wizard confirms that the server meets its prerequisites: it must be a member of an Active Directory domain (or be a Domain Controller itself), and the network must already have active DHCP and DNS servers. DHCP is what supplies IP addresses to PXE clients, and DNS is what resolves server names on the network. On top of this, an NTFS partition must be available to store the boot and install images used during deployment.

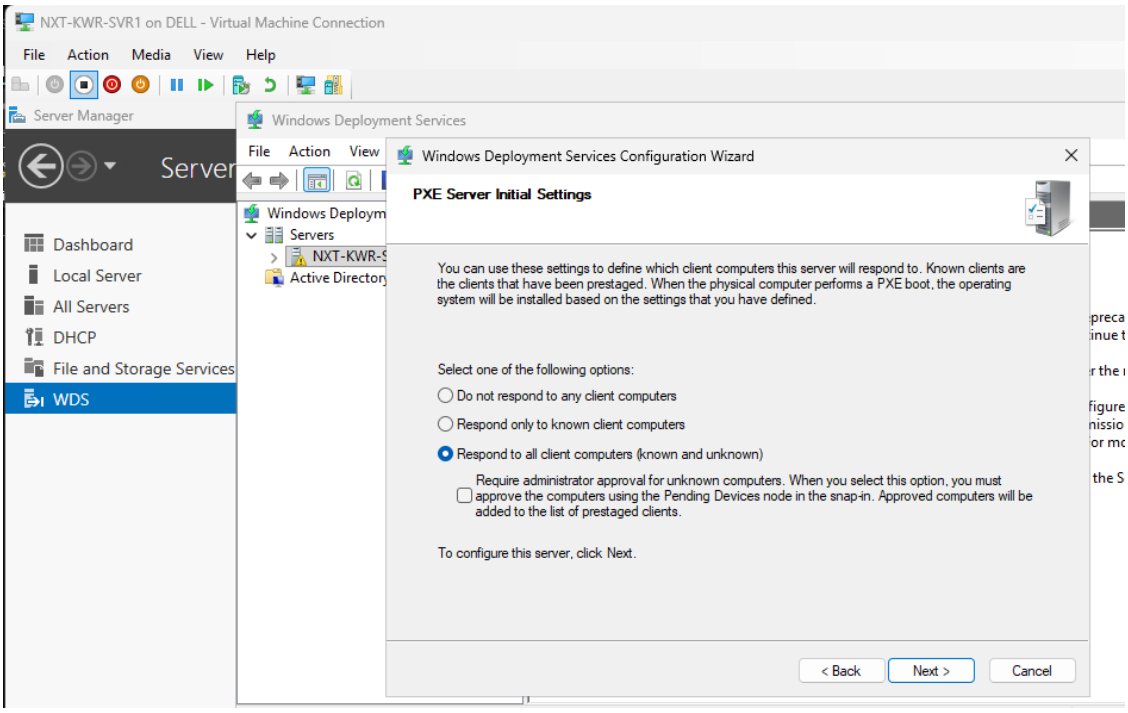


Since this lab is integrated with Active Directory, select the first option and click Next.



Remote Installation Folder

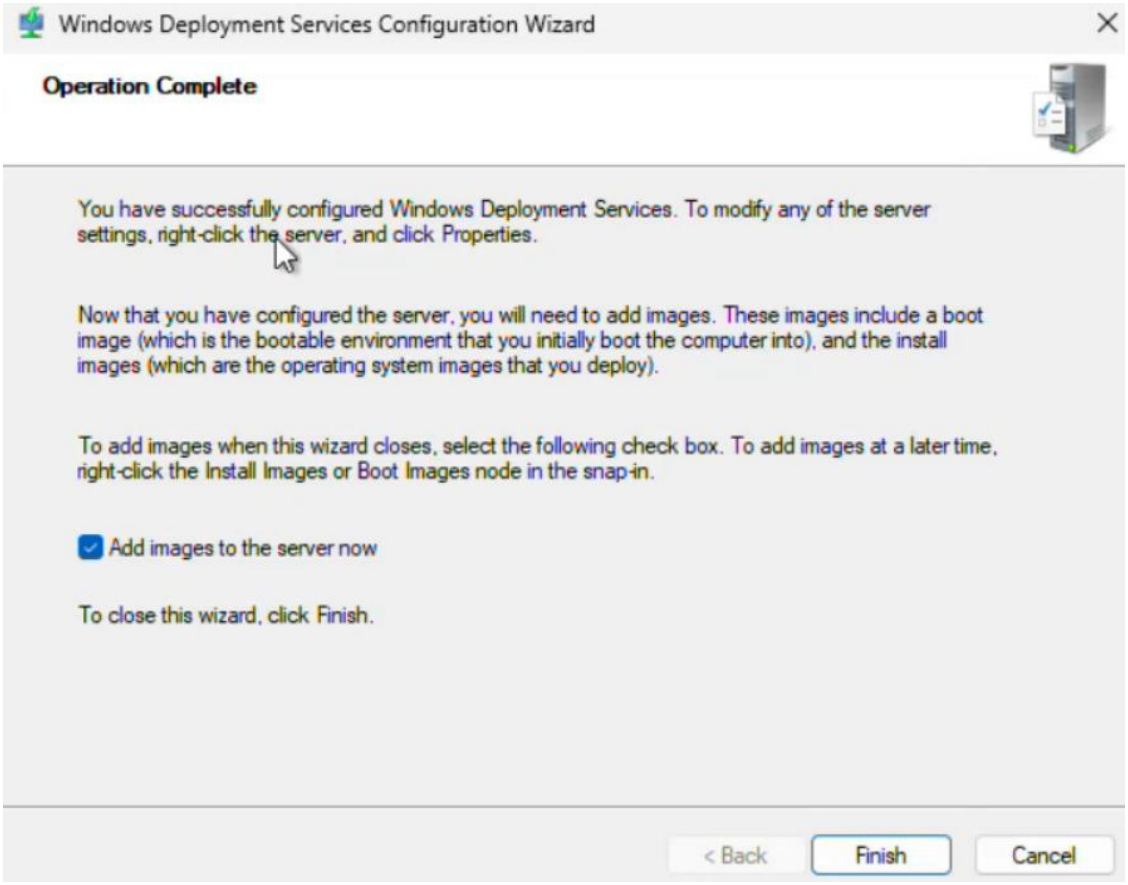
The Remote Installation Folder is where WDS stores its boot images, install images, PXE boot files, and other deployment data. It must sit on an NTFS partition with enough free space. Microsoft recommends using a separate data partition rather than the system partition, both for storage management and to keep system performance unaffected.



PXE Response Settings

This step controls which computers WDS is willing to respond to over PXE. Choosing to respond to all client computers, known and unknown alike, means any device that sends a PXE boot request can connect and begin the deployment process. This is the easiest option and is recommended for testing and lab environments — though it would normally be tightened in a production network.

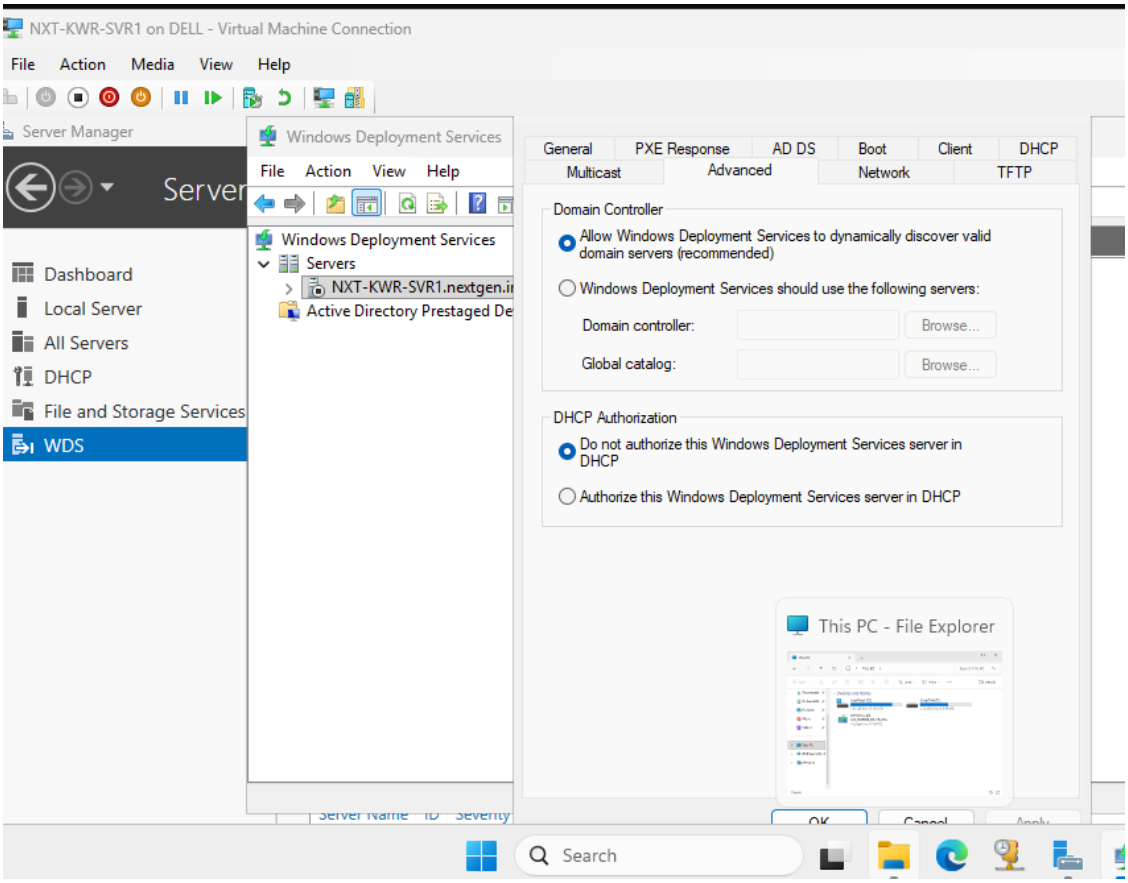
- Known Client — a computer that is already registered in Active Directory.
- Unknown Client — a new computer with no existing registration in Active Directory.
- Administrator Approval (optional) — if enabled, unknown computers must be manually approved by an administrator before deployment can start, adding an extra layer of control in enterprise environments.



Finishing Configuration

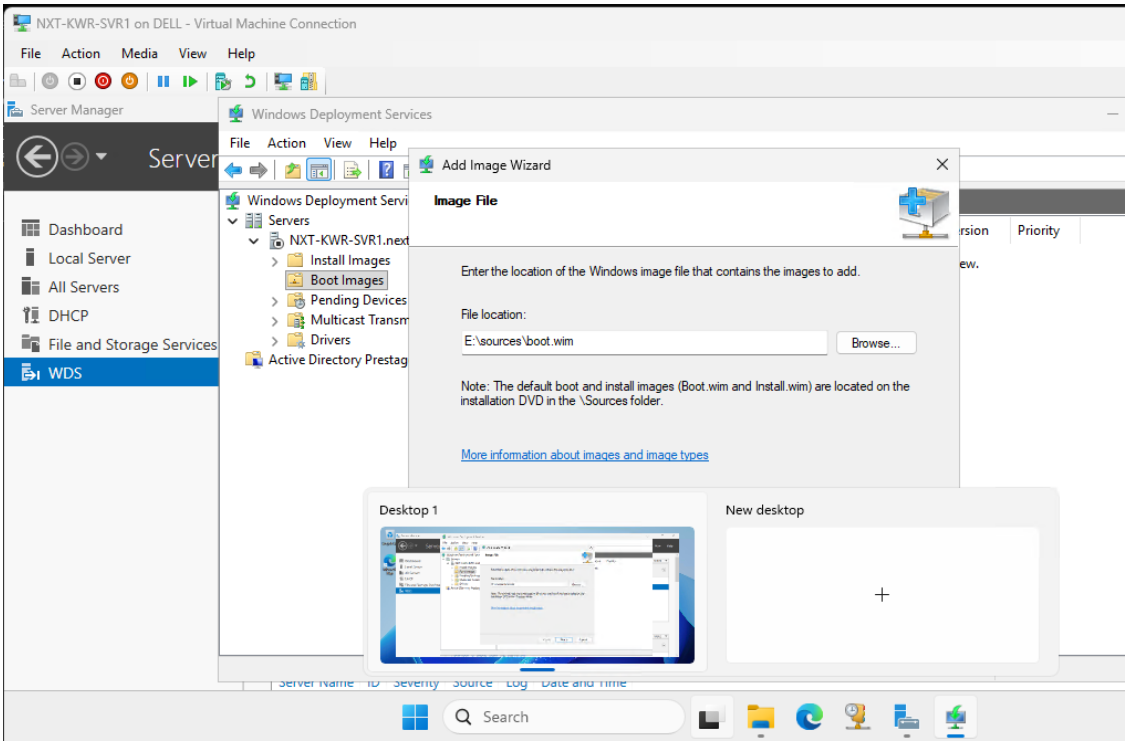
Once the wizard completes, WDS is configured and ready. At this point you are offered the option "Add images to the server now." Since images can be added later, this box can be left unticked here, and Finish clicked to close the wizard. During setup, the RemoteInstall folder was automatically created and shared, ready to hold the images.

Any of these settings can be revisited later by right-clicking the server and choosing Properties.

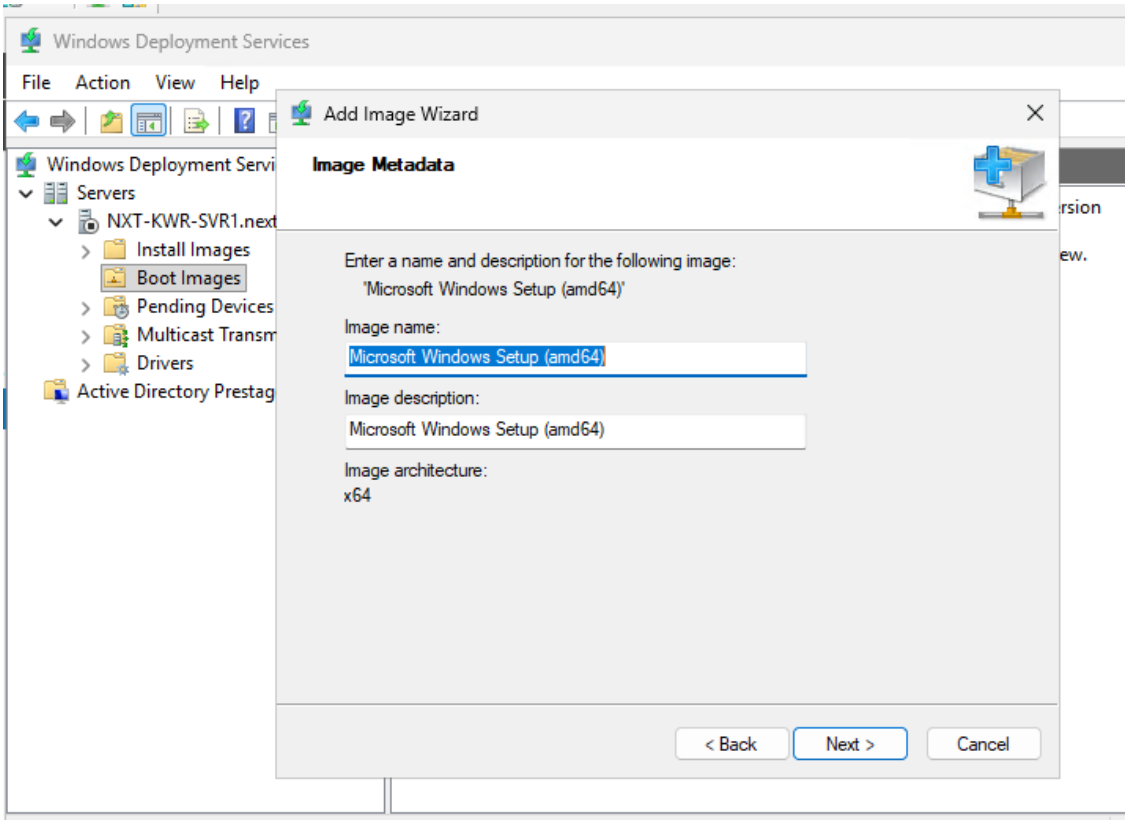


Adding a Boot Image

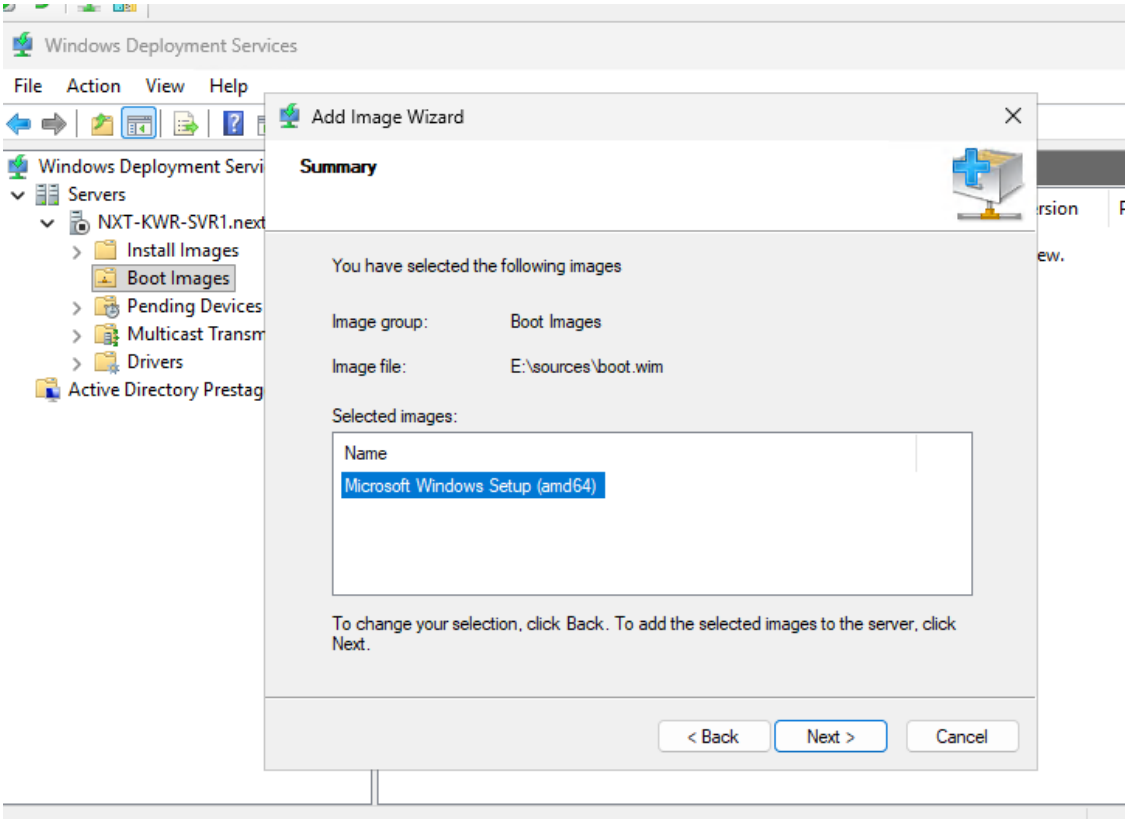
WDS needs two categories of image before it can deploy anything: a Boot Image, which starts the client over PXE and loads the Windows Setup environment, and an Install Image, which contains the actual operating system to be installed. To add the boot image, right-click Boot Images and choose Add Boot Image.



Browse to the path given in the lab instructions to locate the boot.wim file on the installation media.



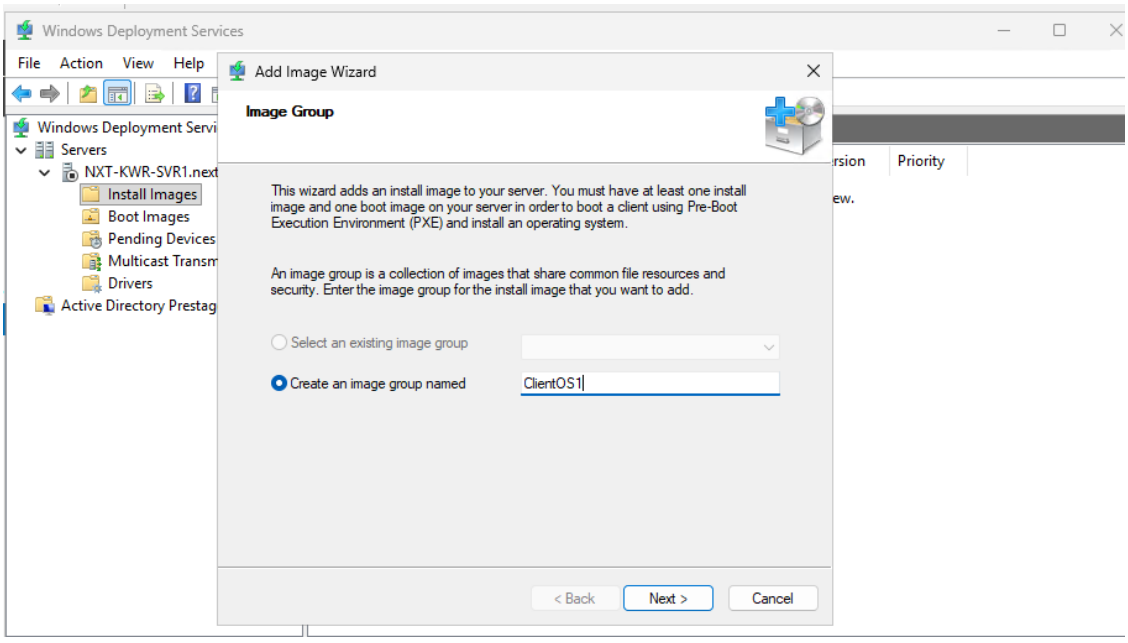
Give the image a clear name that identifies the Windows version — for example, "Windows 10" or "Windows 11" — rather than leaving the default "Microsoft Windows Setup (x64)" label, since you may end up with several boot images over time.



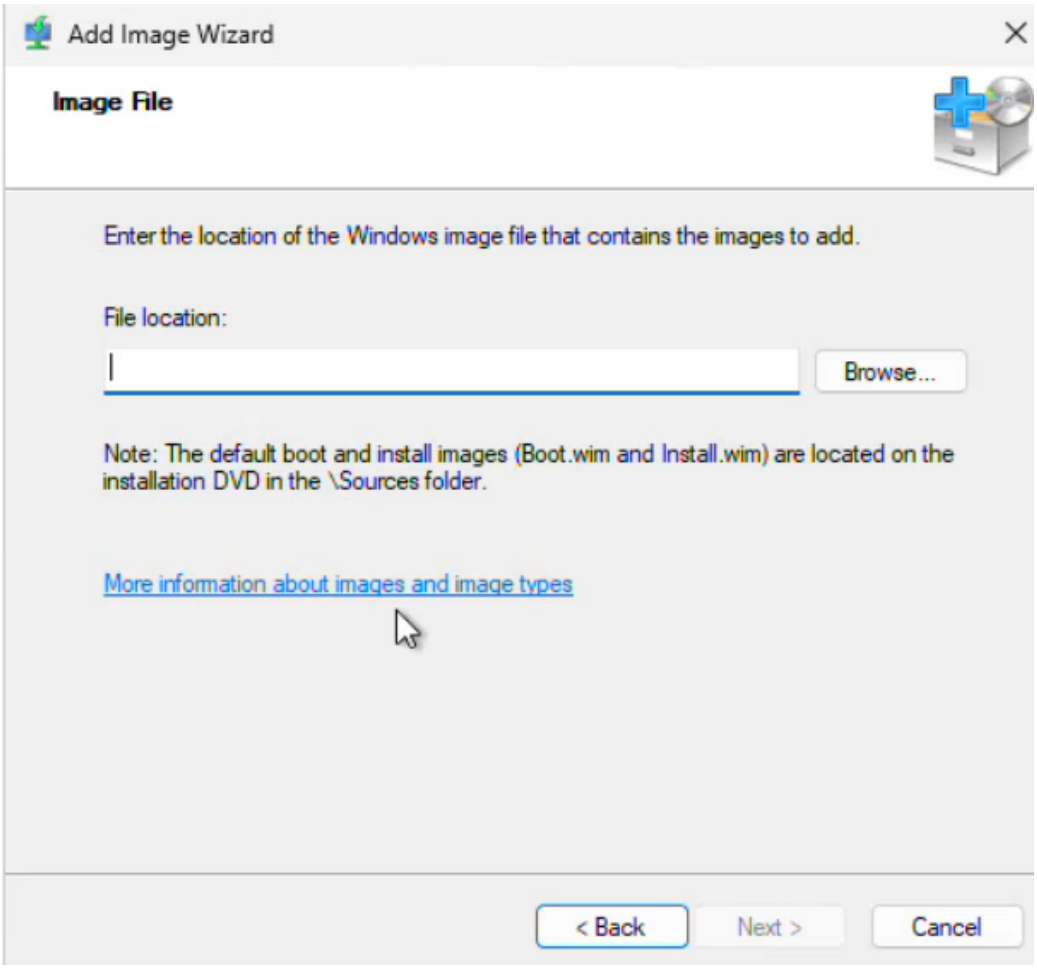
Click Next through the remaining steps to finish adding the boot image.

Adding an Install Image

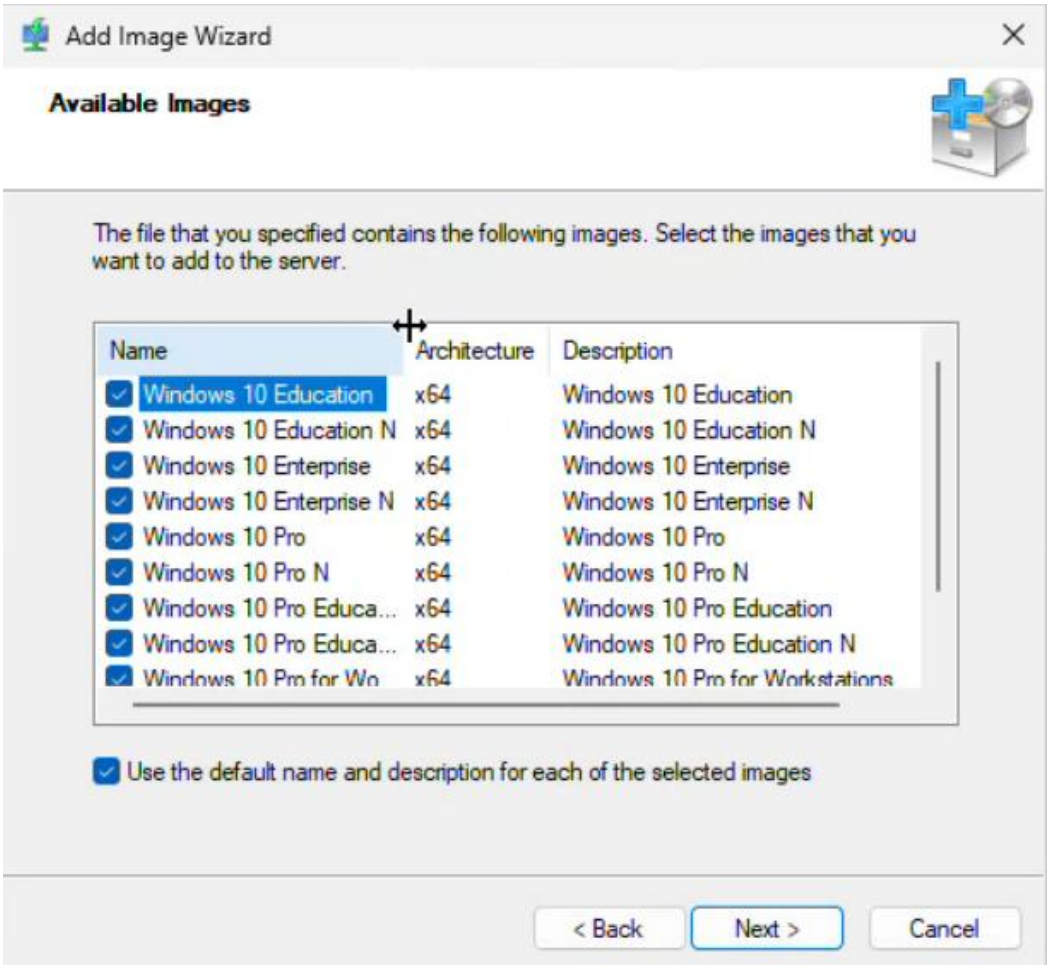
Next, right-click Install Images and choose Add Install Image.



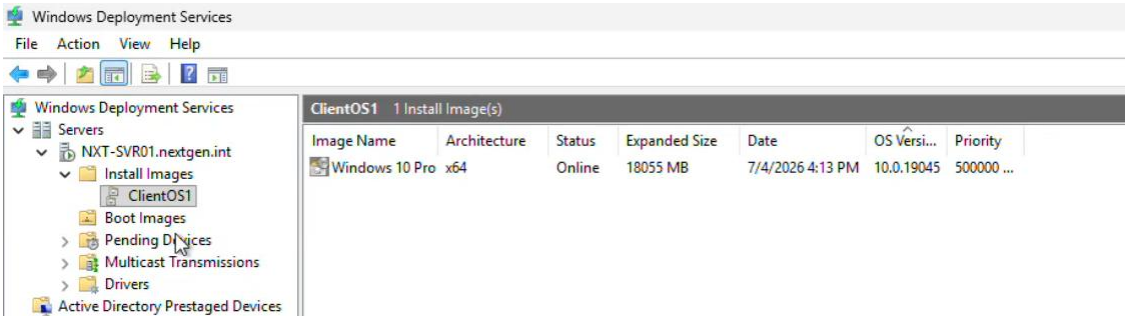
You will first be asked to create or select an Image Group — a folder used to organise install images, useful once you are managing more than one operating system, such as Windows 11 alongside Windows Server.

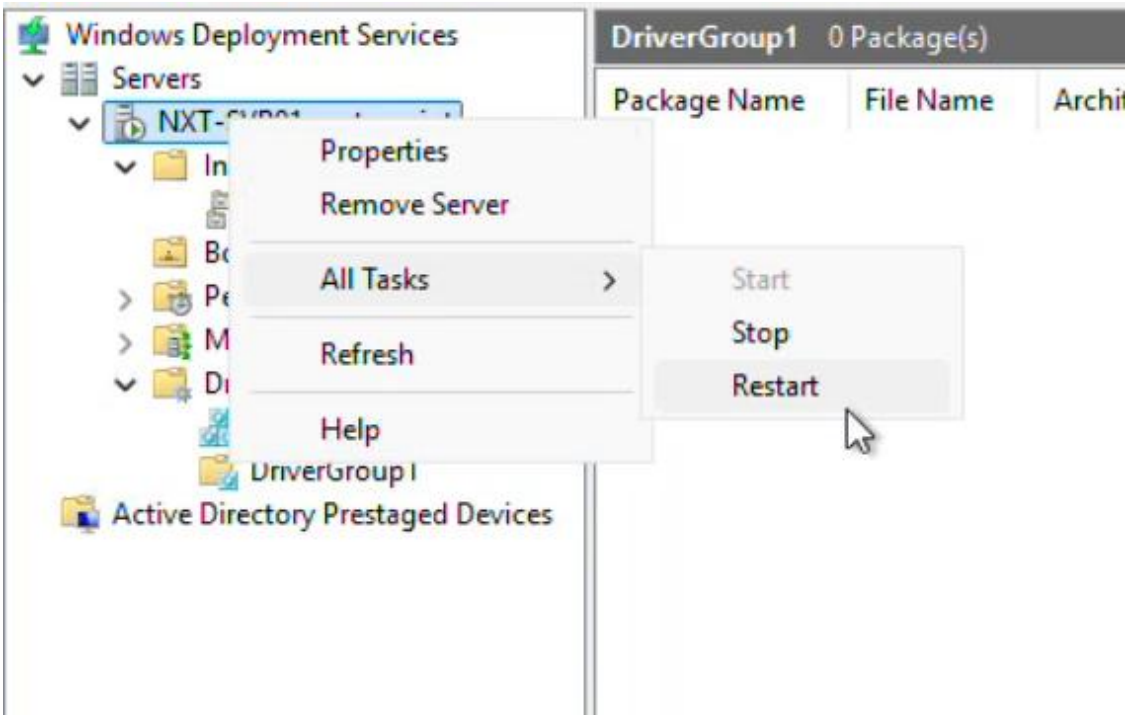


Browse to the install.wim file from the installation media, and choose which edition of the operating system you want to add from the list presented.



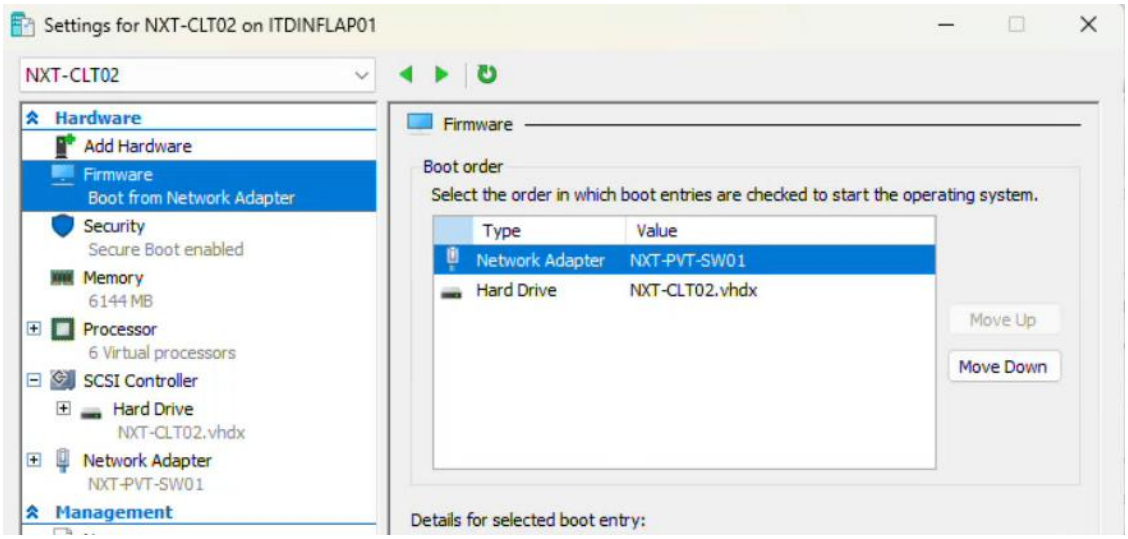
Continue through Next, review the summary screen, and click Finish. Drivers can also be updated through this same server if needed. Once the images are added, restart the WDS server.



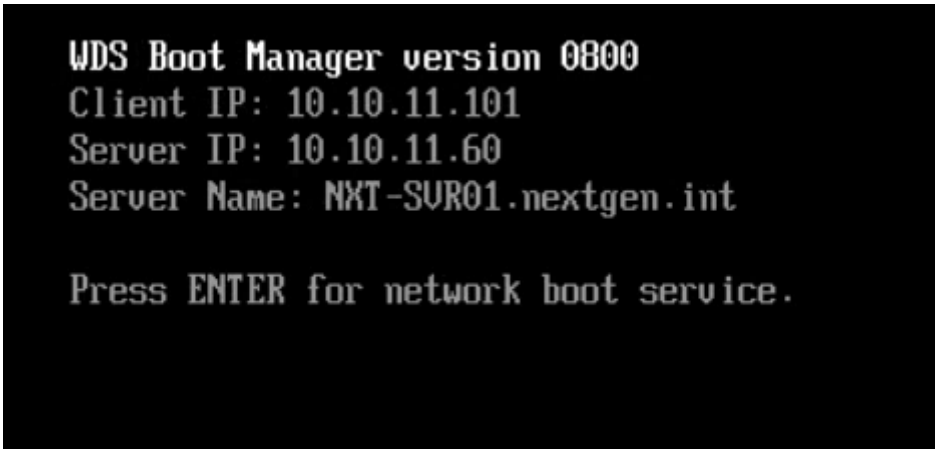


Deploying Windows to a Client

On the client VM, the first boot device needs to be set to Network Adapter so it can PXE boot instead of trying to boot from a local disk.



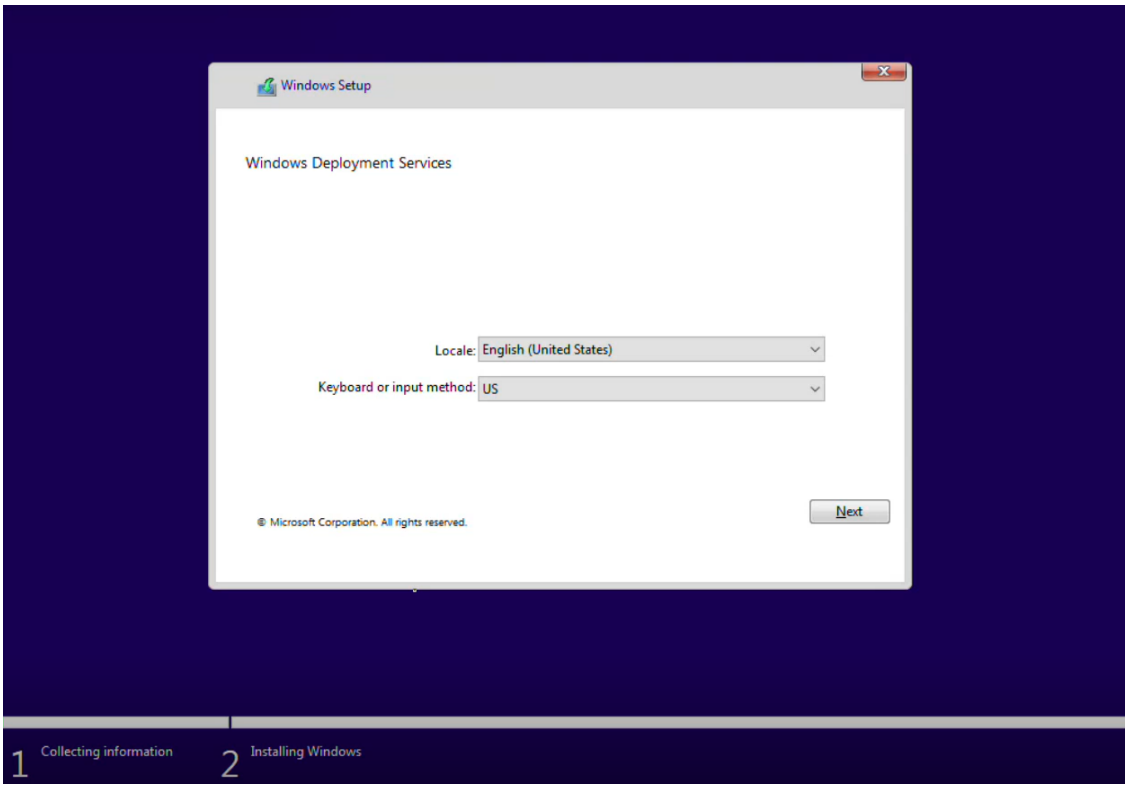
Once connected, the client contacts DHCP and obtains an IP address, just as covered in Part 1.



From there, the client begins booting from the network, downloading the boot image from WDS.



The installation then proceeds much like a normal Windows setup, prompting the administrator to click through several steps.



Automating Installation with unattend.xml

Clicking through the same installation prompts for every single client computer is repetitive and easy to get wrong at scale. An unattend.xml file answers those setup questions automatically —

language, partitioning, product key, and so on — so the installation can run unattended from start to finish.

After Installation: Domain Join and Renaming

Once the WDS-based installation completes, the client computer automatically joins the domain — no manual domain-join step is required afterwards. If the computer's name needs to be changed after installation, this can also be managed centrally through the WDS server, rather than on the client itself.

Summary

DHCP and WDS work as a pair to support automated, network-based deployment. DHCP's DORA process hands out IP addresses and network settings automatically, while its authorization step ensures only a trusted server can do so inside a domain. WDS then builds on that foundation: once a client has an address, it can PXE boot, pull down a boot image, and install a full operating system entirely over the network — with the client joining the domain automatically once setup finishes. Together, these two roles remove almost all of the manual, repetitive work involved in provisioning new computers in a domain environment.